

EXTENSION 2

Stage 6

Calculus (Ext2), C1 Integration (Ext2)

Substitution and Harder Integration

Teacher: Nardene Dodd

Exam Equivalent Time: 51 minutes (based on allocation of 1.5 minutes per mark)



Questions

1. Calculus, EXT2 C2 2020 SPEC2 11 MC

With a suitable substitution $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2(x)}{\sec^2(x) - 3\tan(x) + 1} \, dx$ can be expressed as

- A. $\int_1^{\frac{1}{\sqrt{3}}} \left(\frac{1}{u-1} - \frac{1}{u-2} \right) du$

B. $\int_1^{\sqrt{3}} \left(\frac{1}{u-2} - \frac{1}{u-1} \right) du$

C. $\int_1^{\sqrt{3}} \left(\frac{1}{u-1} - \frac{1}{u-2} \right) du$

D. $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{1}{3(u-1)} - \frac{1}{3(u+2)} \right) du$

2. Calculus, EXT2 C1 2011 HSC 1b

Evaluate $\int_0^3 x\sqrt{x+1} \, dx$. (3 marks)

3. Calculus, EXT2 C1 2019 SPEC1-N 4

Evaluate $\int_{e^3}^{e^4} \frac{1}{x\log_e(x)} \, dx$. (3 marks)

4. Calculus, EXT2 C1 2020 SPEC1 2

Evaluate $\int_{-1}^0 \frac{1+x}{\sqrt{1-x}} \, dx$. (3 marks)

5. Calculus, EXT2 C1 2009 HSC 1a

Find $\int \frac{\ln x}{x} \, dx$. (2 marks)

6. Calculus, EXT2 C1 2010 HSC 1a

Find $\int \frac{x}{\sqrt{1+3x^2}} \, dx$. (2 marks)

7. Calculus, EXT2 C1 2021 HSC 13b

Use an appropriate substitution to evaluate $\int_{\sqrt{10}}^{\sqrt{13}} x^3\sqrt{x^2-9} \, dx$. (3 marks)

8. Calculus, EXT2 C1 2023 HSC 13a

Find $\int \frac{1-x}{\sqrt{5-4x-x^2}} \, dx$. (3 marks)

9. Calculus, EXT2 C1 2007 HSC 1d

Evaluate $\int_0^{\frac{3}{4}} \frac{x}{\sqrt{1-x}} \, dx$. (4 marks)

10. Calculus, EXT2 C1 2009 HSC 1e

Evaluate $\int_1^{\sqrt{3}} \frac{1}{x^2\sqrt{1+x^2}} \, dx$. (4 marks)

11. Calculus, EXT2 C1 2010 HSC 1e

Find $\int \frac{dx}{1+\sqrt{x}}$. (3 marks)

12. Calculus, EXT2 C1 2013 HSC 11d

Evaluate $\int_0^1 x^3\sqrt{1-x^2} \, dx$. (3 marks)

Worked Solutions

1. Calculus, EXT2 C2 2020 SPEC2 11 MC

Let $u = \tan(x)$

$$\frac{du}{dx} = \sec^2(x) \Rightarrow du = \sec^2(x) dx$$

When $x = \frac{\pi}{3}$, $u = \sqrt{3}$

$x = \frac{\pi}{4}$, $u = 1$

$$\begin{aligned} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2(x)}{\sec^2(x) - 3\tan(x) + 1} dx &= \int_1^{\sqrt{3}} \frac{1}{u^2 + 1 - 3u + 1} du \\ &= \int_1^{\sqrt{3}} \frac{1}{u^2 - 3u + 2} du \\ &= \int_1^{\sqrt{3}} \frac{1}{(u-2)(u-1)} du \\ &= \int_1^{\sqrt{3}} \frac{1}{u-2} - \frac{1}{u-1} du \\ &\Rightarrow B \end{aligned}$$

2. Calculus, EXT2 C1 2011 HSC 1b

Let $u = 1 + x$, $du = dx$

When $x = 0$, $u = 1$

$x = 3$, $u = 4$

$$\begin{aligned} \int_0^3 x\sqrt{1+x} dx &= \int_1^4 (u-1)\sqrt{u} du \\ &= \int_1^4 \left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right) du \\ &= \left[\frac{2}{5}u^{\frac{5}{2}} - \frac{2}{3}u^{\frac{3}{2}}\right]_1^4 \\ &= \left(\frac{2}{5} \times 32 - \frac{2}{3} \times 8\right) - \left(\frac{2}{5} - \frac{2}{3}\right) \\ &= \frac{116}{15} \end{aligned}$$

3. Calculus, EXT2 C1 2019 SPEC1-N 4

Let $u = \log_e x$

$$\frac{du}{dx} = \frac{1}{x} \Rightarrow du = \frac{1}{x} dx$$

When $x = e^4 \Rightarrow u = 4$

When $x = e^3 \Rightarrow u = 3$

$$\begin{aligned} \int_{e^3}^{e^4} \frac{1}{x \log_e(x)} dx &= \int_3^4 \frac{1}{u} du \\ &= [\log_e u]_3^4 \\ &= \log_e 4 - \log_e 3 \\ &= \log_e \left(\frac{4}{3}\right) \end{aligned}$$

4. Calculus, EXT2 C1 2020 SPEC1 2

Let $u = 1 - x \Rightarrow x = 1 - u$

$$\frac{du}{dx} = -1 \Rightarrow dx = -du$$

When $x = 0$, $u = 1$

$x = -1$, $u = 2$

$$\begin{aligned} \int_{-1}^0 \frac{1+x}{\sqrt{1-x}} dx &= - \int_2^1 \frac{2-u}{\sqrt{u}} du \\ &= \int_1^2 2u^{-\frac{1}{2}} - u^{\frac{1}{2}} du \\ &= \left[4u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}}\right]_1^2 \\ &= 4\sqrt{2} - \frac{4\sqrt{2}}{3} - \left(4 - \frac{2}{3}\right) \\ &= \frac{8\sqrt{2}}{3} - \frac{10}{3} \end{aligned}$$

5. Calculus, EXT2 C1 2009 HSC 1a

$$\text{Let } u = \ln x, \quad du = \frac{1}{x} dx$$

$$\begin{aligned} \int \frac{\ln x}{x} dx &= \int u du \\ &= \frac{1}{2} u^2 + c \\ &= \frac{1}{2} (\ln x)^2 + c \end{aligned}$$

6. Calculus, EXT2 C1 2010 HSC 1a

$$\text{Let } u = 1 + 3x^2, \quad du = 6x dx$$

$$\begin{aligned} \therefore \int \frac{x}{\sqrt{1+3x^2}} dx &= \frac{1}{6} \int u^{-\frac{1}{2}} du \\ &= \frac{1}{6} \times 2 \times \sqrt{u} + c \\ &= \frac{1}{3} \sqrt{1+3x^2} + c \end{aligned}$$

7. Calculus, EXT2 C1 2021 HSC 13b

$$\int_{\sqrt{10}}^{\sqrt{13}} x^3 \sqrt{x^2-9} dx$$

$$\text{Let } u = x^2 - 9 \Rightarrow x^2 = u + 9$$

$$\frac{du}{dx} = 2x \Rightarrow du = 2x dx$$

$$\text{If } x = \sqrt{13}, \quad u = 4$$

$$\text{If } x = \sqrt{10}, \quad u = 1$$

$$\begin{aligned} \int_{\sqrt{10}}^{\sqrt{13}} x^3 \sqrt{x^2-9} dx &= \int_1^4 (u+9) \sqrt{u} \cdot \frac{1}{2} du \\ &= \frac{1}{2} \int_1^4 u^{\frac{3}{2}} + 9u^{\frac{1}{2}} du \\ &= \frac{1}{2} \left[\frac{2}{5} u^{\frac{5}{2}} + 9 \cdot \frac{2}{3} u^{\frac{3}{2}} \right]_1^4 \\ &= \frac{1}{2} \left[\left(\frac{2}{5} \cdot 4^{\frac{5}{2}} + 6 \cdot 4^{\frac{3}{2}} \right) - \left(\frac{2}{5} + 6 \right) \right] \\ &= \frac{1}{2} \left(\frac{64}{5} + 48 - \frac{32}{5} \right) \\ &= \frac{136}{5} \end{aligned}$$

8. Calculus, EXT2 C1 2023 HSC 13a

$$\begin{aligned} I &= \int \frac{1-x}{\sqrt{5-4x-x^2}} dx \\ &= \frac{1}{2} \int \frac{2-2x}{\sqrt{5-4x-x^2}} dx \\ &= \frac{1}{2} \int \frac{-4-2x}{\sqrt{5-4x-x^2}} dx + \frac{1}{2} \int \frac{6}{\sqrt{5-4x-x^2}} dx \end{aligned}$$

$$\text{Let } u = 5 - 4x - x^2$$

$$\frac{du}{dx} = -4 - 2x \Rightarrow du = -4 - 2x dx$$

$$\begin{aligned} I &= \frac{1}{2} \int u^{-\frac{1}{2}} du + 3 \int \frac{1}{\sqrt{9-(x+2)^2}} dx \\ &= u^{\frac{1}{2}} + 3 \sin^{-1} \left(\frac{x+2}{3} \right) + c \\ &= \sqrt{5-4x-x^2} + 3 \sin^{-1} \left(\frac{x+2}{3} \right) + c \end{aligned}$$

9. Calculus, EXT2 C1 2007 HSC 1d

$$\text{Let } u = 1-x, \quad du = -dx, \quad x = 1-u$$

$$\text{When } x = 0, \quad u = 1$$

$$\text{When } x = \frac{3}{4}, \quad u = \frac{1}{4}$$

$$\begin{aligned} \int_0^{\frac{3}{4}} \frac{x dx}{\sqrt{1-x}} &= \int_1^{\frac{1}{4}} \frac{(1-u)(-du)}{\sqrt{u}} \\ &= \int_{\frac{1}{4}}^1 \left(u^{-\frac{1}{2}} - u^{\frac{1}{2}} \right) du \\ &= \left[2u^{\frac{1}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_{\frac{1}{4}}^1 \\ &= \left[\left(2 - \frac{2}{3} \right) - \left(1 - \frac{1}{12} \right) \right] \\ &= \frac{5}{12} \end{aligned}$$

10. Calculus, EXT2 C1 2009 HSC 1e

Let $x = \tan\theta$, $dx = \sec^2\theta d\theta$

When $x = 1$, $\theta = \frac{\pi}{4}$

When $x = \sqrt{3}$, $\theta = \frac{\pi}{3}$

$$\begin{aligned}\int_1^{\sqrt{3}} \frac{1}{x^2\sqrt{1+x^2}} dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2\theta}{\tan^2\theta\sqrt{1+\tan^2\theta}} d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2\theta}{\tan^2\theta\sec\theta} d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec\theta}{\tan^2\theta} d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos^2\theta}{\cos\theta\sin^2\theta} d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos\theta d\theta}{\sin^2\theta} \\ &= \left[-\frac{1}{\sin\theta} \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \\ &= \left(-\frac{2}{\sqrt{3}} + \sqrt{2} \right) \\ &= \sqrt{2} - \frac{2\sqrt{3}}{3}\end{aligned}$$

11. Calculus, EXT2 C1 2010 HSC 1e

Solution 1

Let $u = \sqrt{x}$, $du = \frac{1}{2\sqrt{x}} dx$, $dx = 2u du$

$$\begin{aligned}\int \frac{dx}{1+\sqrt{x}} &= \int \frac{2u du}{1+u} \\ &= \int \left(\frac{2+2u-2}{1+u} \right) du \\ &= \int \left(2 - \frac{2}{1+u} \right) du \\ &= 2u - 2\log_e(1+u) + c_1 \\ &= 2\sqrt{x} - 2\log_e(1+\sqrt{x}) + c_1\end{aligned}$$

Alternative Solution

Let $u = 1 + \sqrt{x}$, $du = \frac{1}{2\sqrt{x}} dx$, $dx = 2(u-1) du$

$$\begin{aligned}\int \frac{dx}{1+\sqrt{x}} &= 2 \int \frac{u-1}{u} du \\ &= 2 \int \left(1 - \frac{1}{u} \right) du \\ &= 2u - 2\log_e |u| + c_2 \\ &= 2 + 2\sqrt{x} - 2\log_e(1+\sqrt{x}) + c_2\end{aligned}$$

NB. These solutions are equivalent by making $c_1 = c_2 + 2$

12. Calculus, EXT2 C1 2013 HSC 11d

Let $u = 1 - x^2$, $du = -2x \, dx$

When $x = 0$, $u = 1$

When $x = 1$, $u = 0$

$$\begin{aligned} \int_0^1 x^3 \sqrt{1-x^2} \, dx &= \int_1^0 x^2 \sqrt{1-x^2} x \, dx \\ &= \int_1^0 (1-u) \sqrt{u} \times \left(\frac{-du}{2} \right) \\ &= \frac{1}{2} \int_0^1 \left(u^{\frac{1}{2}} - u^{\frac{3}{2}} \right) du \\ &= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} - \frac{2}{5} u^{\frac{5}{2}} \right]_1^0 \\ &= \frac{1}{2} \left[\left(\frac{2}{3} - \frac{2}{5} \right) - 0 \right] \\ &= \frac{2}{15} \end{aligned}$$

STRATEGY: The choice of $u = 1 - x^2$ or $u^2 = 1 - x^2$ provided much less calculation than $u = x^2$. Take note!